

```
1  #include <cs50.h>
2  #include <stdio.h>
3  #include <stdlib.h>
4  #include <string.h>
5
6  char *concat(char *s1, char *s2);
7
8  int main(void)
9  {
10     char *s1 = get_string("String 1: ");
11     char *s2 = get_string("String 2: ");
12
13     char *s3 = concat(s1, s2);
14     printf("%s\n", s3);
15     free(s3);
16 }
17
18 char *concat(char *s1, char *s2)
19 {
20     // Get memory for s3
21     int len1 = strlen(s1);
22     int len2 = strlen(s2);
23     char *s3 = malloc(len1 + len2 + 1);
24
25     // Copy over the values from s1 and s2
26     for (int i = 0; i < len1; i++)
27     {
28         s3[i] = s1[i];
29     }
30     for (int i = 0; i < len2; i++)
31     {
32         s3[len1 + i] = s2[i];
33     }
34
35     s3[len1 + len2] = '\0';
36     return s3;
37 }
```

```
1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <string.h>
4
5  int main(int argc, char *argv[])
6  {
7      // Check for improper usage, otherwise, get filename length
8      if (argc != 2)
9      {
10         printf("Wrong usage: Try ./create [filename]\n");
11         return 1;
12     }
13     int filename_length = strlen(argv[1]);
14
15     // Create a new block of memory to store filename
16     char *filename = malloc(sizeof(char) * (filename_length + 1));
17
18     // Check if malloc failed
19     if (filename == NULL)
20     {
21         printf("Malloc failed.");
22         return 1;
23     }
24
25     // Copy argv[1] into block of memory for filename
26     sprintf(filename, "%s", argv[1]);
27
28     // Open new file under the name stored at filename
29     FILE *new_file = fopen(filename, "w");
30
31     // Check if fopen failed
32     if (new_file == NULL)
33     {
34         printf("Could not create file.");
35         return 1;
36     }
37
38     free(filename);
39     fclose(new_file);
40 }
```

```
1  #include <cs50.h>
2  #include <stdint.h>
3  #include <stdio.h>
4
5  int main(int argc, char *argv[])
6  {
7      // Check for improper usage
8      if (argc != 2)
9      {
10         printf("Improper usage.\n");
11         return 1;
12     }
13
14     // Open PDF with inputted filename
15     FILE *pdf = fopen(argv[1], "r");
16
17     uint8_t buffer[4];
18     uint8_t signature[] = {0x25, 0x50, 0x44, 0x46};
19
20     // Read the first 4 bytes into the buffer
21     fread(buffer, 1, 4, pdf);
22     fclose(pdf);
23
24     // Check the buffer contents against the PDF signature
25     for (int i = 0; i < 4; i++)
26     {
27         if (buffer[i] != signature[i])
28         {
29             printf("Not a PDF.\n");
30             return 0;
31         }
32     }
33     printf("Likely a PDF!");
34     return 0;
35 }
```